



Control Panel Sequence of Operations Altronic Control Panel 5900-FLX36-V2-014 4-throw, 3 stage Ariel with ASC

Disclaimer

This Sequence of Operations (SOO) describes the general functionality and use of the Altronic, LLC control panel, including its operation with the DE-4000 Controller. This document is intended to provide a high-level overview of system behavior and is not a substitute for proper engineering review, safety analysis, or site-specific implementation details.

Altronic, LLC makes no guarantees regarding the suitability, accuracy, or completeness of this SOO for any specific application. The DE-4000 Controller is a configurable system, and its operation depends on user-defined settings, logic, and parameters. As a result, Altronic, LLC cannot guarantee system performance under all configurations or applications. The use of this document and associated systems is at the sole risk of the user. Altronic, LLC is not responsible for any damages, malfunctions, or failures resulting from improper use, installation, modification, or misinterpretation of this document. Users should verify all control logic and safety protocols before commissioning.

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Revision History

[illegible]

System Overview

The Altronic, LLC DE-4000 Series is a configurable Safety Shutdown and Control System designed to monitor, control, and protect critical rotating machinery. This microprocessor-based system processes both analog and digital inputs to execute automated start-up and shutdown sequences. Additional functions include pre-lube, engine warm-up, and cool-down operations.

The DE-4000 control panel includes a touchscreen HMI (Human-Machine Interface), which allows operators to start/stop equipment, adjust process setpoints, and monitor system status. Configuration of the DE-4000 is performed via the onboard programming interface, allowing users to tailor control sequences to specific application needs.

Support Documentation

This Sequence of Operations (SOO) is specific to the **Altronic, LLC DE-4000 Control Panel** and describes its operational functionality. For additional information, installation guidelines, configuration details, system drawings, and firmware updates, please refer to the following official documentation:

- **DE-4000 Product Page:**
<https://www.altronic-llc.com/product/controls/de-4000/>
- **DE-4000 Brochure:**
<https://www.altronic-llc.com/wp-content/uploads/DE-4000-Brochure.pdf>
- **DE-4000 System Installation Instructions:**
<https://www.altronic-llc.com/documentation/de-4000-system-installation-instructions/>
- **DE-4000 Operation and Configuration Manual:**
<https://www.altronic-llc.com/documentation/de-4000/operation-and-configuration-manual/>
- **DE-4000 Firmware Download Quick Start Guide:**
<https://www.altronic-llc.com/wp-content/uploads/DE-4000-Firmware-Download-Quick-Start-Guide.pdf>
- **System Drawings:** *(Refer to project-specific system drawings for wiring, connections, and installation details.)*

Users should refer to these resources for detailed system specifications, installation procedures, system drawings, and firmware updates. This SOO document focuses specifically on the operation of the **DE-4000 Control Panel** and should be used in conjunction with official documentation to ensure proper configuration and usage.

DE-4000 Shutdown List

The **DE-4000 Shutdown List** provides a comprehensive overview of system fault conditions and their classifications. These faults are configured within the **DE-4000 Control Panel** to monitor critical parameters and trigger appropriate shutdown actions when necessary. The following tables detail these fault conditions, their settings, and response classifications.

Selectable Fault Types

Each fault in the DE-4000 system is categorized into one of the following **fault classes**, determining how and when the fault becomes active:

- **A** – Armed immediately
- **B** – Selectable time delay at reset before becoming armed
- **C** – Armed after the fault clears with a selectable time delay
- **S** – Scripted logic-based fault activation

These classifications ensure flexibility in system response, allowing operators to fine-tune safety and operational settings as needed.

Terminal Board 1

Channel	Description	I/O Type	Range	Unit of Measure	HIGH SD Class If-B/C SECS		HIGH Shutdown Setpoint	LOW or DI SD Class (If - B) SECS		LOW Shutdown Setpoint	HIGH Alarm Setpoint	LOW Alarm Setpoint	Modbus Reg 16 Bit (Node # set in DE4000)	Modbus Reg 32 Bit (Node 200)
IN1	SUCTION INLET PRESSURE	0-5VDC (ANALOG)	0-300	psig	N/A			N/A					40090	40001
IN2	FIELD PRESSURE	0-5VDC (ANALOG)	0-300	psig	N/A			N/A					40091	40003
IN3	ENGINE VSHH	N/C DI						A					40092	40005
IN4	COOLER 1 VSHH	N/C DI						A					40093	40007
IN5	COOLER 2 VSHH	N/C DI						A					40094	40009
IN6	SCRUBBER 1 VSHH	N/C DI						A					40095	40011
IN7	SCRUBBER 2 VSHH	N/C DI						A					40096	40013
IN8	SCRUBBER 3 VSHH	N/C DI						A					40097	40015
IN9	FUEL SCRUBBER LSHH	N/C DI						A					40098	40017
IN10	ENGINE OIL LSL	N/C DI						A					40099	40019
IN11	COMP OIL LSL	N/C DI						A					40100	40021
IN12	JACKET WATER LSL	N/C DI						A					40101	40023
IN13	AUX WATER LSL	N/C DI						A					40102	40025
IN14	ASC ALARM	N/C DI						B	30				40103	40027
IN15	ASC FAULT	N/C DI						B	30				40104	40029
IN16	CUSTOMER REMOTE E-STOP	N/C DI						A					40105	40031
IN17		NOT USED											40106	40033
IN18		NOT USED											40107	40035
IN19		NOT USED											40108	40037
IN20		NOT USED											40109	40039
IN21		NOT USED											40110	40041
IN22		NOT USED											40111	40043
IN23		NOT USED											40112	40045
IN24		NOT USED											40113	40047
IN25		NOT USED											40114	40049
IN26		NOT USED											40115	40051
IN27		NOT USED											40116	40053
IN28		NOT USED											40117	40055
IN29		NOT USED											40118	40057
IN30	CUSTOMER REMOTE STOP	N/C DI						A					40119	40059
IN31	ENG/MOTOR FAULT	N/C DI						B	30				40120	40061
IN32	PANEL EMERGENCY STOP	N/C DI						A					40121	40063
AO1	SUCTION CONTROL												40260	-
AO2	DRIVER SPEED CONTROL												40261	-
AO3	RECYCLE CONTROL												40262	-
AO4													40263	-
DO1	FUEL GAS VALVE												40020 Bit 0	-
DO2	RATED RPM PERMISSIVE												40020 Bit 1	-
DO3	ASC RESET												40020 Bit 2	-
DO4	PROPANE COLD START												40020 Bit 3	-
DO5	BATTERY SAVER												40020 Bit 4	-
DO6	BATTERY SAVER (FUTURE)												40020 Bit 5	-
DO7	BATTERY SAVER (FUTURE)												40020 Bit 6	-
DO8	STATUS												40020 Bit 7	-
RPM1	ENGINE SPEED												40250	-
RPM2													40251	-

**Additional Modbus Register List available that includes DE4000 Standard Modbus Registers upon request.*

Terminal Board 2

Channel	Description	I/O Type	Range	Unit of Measure	HIGH SD Class (If - B) SECs		HIGH Shutdown Setpoint	LOW or DI SD Class (If - B) SECs		LOW Shutdown Setpoint	HIGH Alarm Setpoint	LOW Alarm Setpoint	Modbus Reg 16 Bit (Node # set in DE4000)	Modbus Reg 32 Bit (Node 200)
IN1	ENGINE AUX WATER	J/K T/C	0-1200	°F	N/A			N/A					40122	40065
IN2	1ST INTERSTAGE COOLER	J/K T/C	0-1200	°F	N/A			N/A					40123	40067
IN3	2nd INTERSTAGE COOLER	J/K T/C	0-1200	°F	N/A			N/A					40124	40069
IN4	3RD STAGE DISCHARGE	J/K T/C	0-1200	°F	N/A			N/A					40125	40071
IN5	LOCAL LOCKOUT	N/C DI						A					40126	40073
IN6	BLOWDOWN OPENED	N/C DI						N/A					40127	40075
IN7	BLOWDOWN CLOSED	N/C DI						N/A					40128	40077
IN8	BYPASS VALVE OPENED	N/C DI						N/A					40129	40079
IN9	BYPASS VALVE CLOSED	N/C DI						N/A					40130	40081
IN10		NOT USED											40131	40083
IN11		NOT USED											40132	40085
IN12		NOT USED											40133	40087
IN13		NOT USED											40134	40089
IN14		NOT USED											40135	40091
IN15		NOT USED											40136	40093
IN16		NOT USED											40137	40095
IN17		NOT USED											40138	40097
IN18		NOT USED											40139	40099
IN19		NOT USED											40140	40101
IN20		NOT USED											40141	40103
IN21		NOT USED											40142	40105
IN22		NOT USED											40143	40107
IN23		NOT USED											40144	40109
IN24		NOT USED											40145	40111
IN25		NOT USED											40146	40113
IN26		NOT USED											40147	40115
IN27		NOT USED											40148	40117
IN28		NOT USED											40149	40119
IN29		NOT USED											40150	40121
IN30		NOT USED											40151	40123
IN31		NOT USED											40152	40125
IN32		NOT USED											40153	40127
AO1	LOUVER 1												40264	-
AO2	LOUVER 2												40265	-
AO3	LOUVER 3												40266	-
AO4	LOUVER 4												40267	-
DO1	BLOWDOWN VALVE												40021 Bit 0	-
DO2	BYPASS VALVE												40021 Bit 1	-
DO3													40021 Bit 2	-
DO4													40021 Bit 3	-
DO5													40021 Bit 4	-
DO6													40021 Bit 5	-
DO7													40021 Bit 6	-
DO8													40021 Bit 7	-
RPM1													40252	-
RPM2													40253	-

**Additional Modbus Register List available that includes DE4000 Standard Modbus Registers upon request.*

System Start-Up Sequence

The Start-Up menu allows users to configure an auto start sequence with a high degree of flexibility. The system progresses through ten sequential states (State 0 to State 9), with transitions controlled by configured permissives and timers. The Start Button initiates the sequence.

Only the DS Outputs are configurable within the Start-Up Sequence. The Terminal Board Digital Outputs (DO) should be configured separately via the Outputs page or through scripting. Refer to the **DE-4000 Operation and Configuration Manual** for configuring Digital Outputs. The setup and control of DOs within this system can be found in the **Digital Outputs** section of this document.

To bypass a state, set its timers to 0 and disable the permissives by unchecking the Enable checkbox.

For specific details, including timing parameters and configuration options, refer to the DE-4000 Operation and Configuration Manual.

State 0

Ready to Start

Initial state. When all faults are cleared, the Start Button can be pressed to begin the sequence.

Additional Notes
If the final discharge pressure is greater than or equal to the setpoint defined in the Global Parameters, then the unit will go through a blowdown cycle.
When an Auto Start is initiated, the unit should go through a warning state (State 1).
When a Local Start is initiated (started on site by the operator at the unit) the Warning state will be skipped.
When the personnel onsite lockout is active, the Panel Light will be set to solid yellow and the Horn will be off. If personnel lockout is not active, the Panel Light will be solid red and the horn will be off.
Auto Start/Stop description of configuration and operation, is at the end of this document. Personnel Lockout switch input will be used as the physical input for the Auto Start configuration.

State 1

Label	Warning	Timer:			min		sec	
		Timeout:			min		sec	
DS Outputs								
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	☐	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	☐	DS4
Permissive A								
☒	Enable	Channel				REL	Setpoint	
		Virtual				=	199	
		Channel Name: Master_State						
Permissive B								
☐	Enable	Channel				REL	Setpoint	
		Channel Name:						
Additional Notes								
Panel Light will be set to Red Strobe and the horn will be ON. Time duration of the Warning State is determined by the global parameter "WarningTimer." This State will be skipped if the System is started locally.								

State 2

Label	Purge/BD	Timer:			min		sec
		Timeout:			min		sec
DS Outputs							
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	☒ DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	☒ DS4
Permissive A							
☒	Enable	Channel				REL	Setpoint
		Virtual				=	299
		Channel Name: Master_State					
Permissive B							
☐	Enable	Channel				REL	Setpoint
		Channel Name:					
Additional Notes							
The Panel Light and Horn will be determined by the previous state. If the Warning State occurred the Panel Light will be red strobe and the Horn will be on. If the Warning State was not required the Panel Light will be Red Strobe and the Horn will be off. Valve positions can be confirmed manually via push buttons on Dashboard Page 2 at the appropriate time in the purge or blowdown cycle. If the valve outputs are "Disabled" on the HMI Dashboard, positions must be confirmed via these push buttons to proceed with Purge/Blowdown. This State will be skipped if a Purge/Blowdown Cycle is not required.							

State 2 Additional Notes (continued)

PURGE CYCLE

- 1) If a Purge is not required the controller will advance to the BLOWDOWN CYCLE.
- 2) If a Purge is required the following will occur:
 - a. If the unit was started in Local Mode, then the status display will read "SCT VLV MUST REMAIN OPEN DURING STARTING". **This is a reminder to the operator to open the Suction Valve.**
 - i. After 15 seconds the unit will advance.
 - b. The Blowdown Valve output (if enabled) is commanded to open. The status display will read **"UNIT PURGE - OPENING BLOWDOWN VALVE"**.
 - c. Once the Blowdown Valve position is confirmed open (via limit switch or manual acknowledgement), the Recycle Valve (if enabled) will be commanded closed.
 - i. **If the Blowdown Valve is Enabled, then the unit will wait for**
 - ii. **If the Blowdown Valve is Disabled; a purge cycle will still**
 - d. Once the Recycle Valve position is confirmed closed, the Bypass Valve output (if enabled) will remain commanded closed.
 - e. Once the Bypass Valve position is confirmed closed (via limit switch or manual acknowledgement), the 'Purge with Bypass Valve Closed Delay' will begin counting down. The status display will read "PURGE W/BP CLOSED COMP IN XXX SECONDS".
 - f. When the Purge with Bypass Closed Delay expires, the Recycle Valve (if enabled) is commanded to open.
 - g. Once the Recycle Valve is confirmed open, the Bypass Valve output (if enabled) is commanded to open. The Bypass Valve output (if enabled) is energized to fully open the valve.
 - h. Once the Bypass Valve is confirmed open (via limit switch or manual acknowledgement), the Purge with Bypass Open Delay will begin timing. The status display will read "PURGE W/BP OPEN COMPLETE IN XXX SECONDS".
 - i. When the Purge with Bypass Open Delay expires, the Blowdown Valve output (if enabled) is commanded closed. The status display will read "CLOSING BD/BD FTC-SD IN XXX SECONDS".
 - j. Once the Blowdown Valve position is confirmed closed (via limit switches or manual acknowledgement), the Purge Permissive Delay will begin timing. The status display will read "NEED SUCT/FAIL TO PURGE SD IN XXX SECONDS".

- k. The controller will monitor and compare the 1st Stage Suction Pressure against the Suction Pressure – Purge Permissive setpoint. If the 1st Stage Suction Pressure does not reach the Suction Pressure – Purge Permissive setpoint before the Purge Permissive Delay expires, a fault-shutdown will occur on Purge Permissive Failure.
- 3) If the 1st Stage Suction Pressure is equal or above the Suction Pressure – Purge Permissive setpoint, the controller advances to the BLOWDOWN CYCLE.

BLOWDOWN CYCLE

- 1) The Final Discharge Pressure will be monitored and compared to the 'Blowdown Required Pressure' setpoint for Blowdown of the unit. If the Final Discharge Pressure is below the 'Blowdown Required Pressure' set point, the Blowdown cycle will be skipped.
- 2) If Final Discharge Pressure is equal to or above the 'Blowdown Required Pressure' setpoint, the Blowdown cycle will be performed.
- 3) The Blowdown Valve output (if enabled) is commanded open. The status display will read "UNIT BLOWDOWN - OPENING BLOWDOWN VALVE"
- 4) Once the Blowdown Valve position is confirmed open (via limit switch or manual acknowledgement), the controller will monitor and compare the 3rd Stage Discharge Pressure against the 'Blowdown Required Pressure' setpoint.
- 5) Once the Final Discharge Pressure is below the 'Blowdown Required Pressure' setpoint the Blowdown cycle is complete and the Blowdown Valve output (if enabled) commanded close.
- 6) Once the Blowdown Valve position is confirmed closed (via limit switch or manual acknowledgement), the controller will advance.

State 3

Label	Pre-Lube	Timer:			min		sec	
		Timeout:			min		sec	
DS Outputs								
Output On (Beginning of State)	☐	DS1	☐	DS2	☒	DS3	☒	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	☒	DS4
Permissive A								
☒	Enable	Channel				REL	Setpoint	
		Virtual				>	10	
		Channel Name: COP						
Permissive B								
☐	Enable	Channel				REL	Setpoint	
		Channel Name:						
Additional Notes								
The Panel Light and Horn will be OFF. Permissive: Comp Oil Pressure								

State 4

Label	DER	Timer:		min	3	sec		
		Timeout:		min		sec		
DS Outputs								
Output On (Beginning of State)	☒	DS1	☐	DS2	☒	DS3	☐	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	☐	DS4
Permissive A								
☐	Enable	Channel				REL	Setpoint	
		Channel Name:						
Permissive B								
☐	Enable	Channel				REL	Setpoint	
		Channel Name:						
Additional Notes								
The Panel Light and Horn will be OFF.								

State 5

Label	Start Initiate	Timer:			min		sec	
		Timeout:			min		sec	
DS Outputs								
Output On (Beginning of State)	☐	DS1	☒	DS2	☐	DS3	☐	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	☐	DS4
Permissive A								
☐	Enable	Channel				REL	Setpoint	
		Channel Name:						
Permissive B								
☐	Enable	Channel				REL	Setpoint	
		Channel Name:						
Additional Notes								
The Panel Light and Horn will be OFF.								

State 6

Label	Crank Wait	Timer:			min		sec	
		Timeout:			min	20	sec	
DS Outputs								
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	☐	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☒	DS3	☐	DS4
Permissive A								
☒	Enable	Channel				REL	Setpoint	
		Terminal Board	1	RPM		>	400	
		Channel Name:		Engine RPM				
Permissive B								
☐	Enable	Channel				REL	Setpoint	
		Channel Name:						
Additional Notes								
The Panel Light and Horn will be OFF. Permissive: Engine RPM								

State 7

Label	Warmup	Timer:			min	60	sec	
		Timeout:			min		sec	
DS Outputs								
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	☐	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	☐	DS4
Permissive A								
☒	Enable	Channel				REL	Setpoint	
		Virtual				>	80	
		Channel Name: COT						
Permissive B								
☒	Enable	Channel				REL	Setpoint	
		Virtual				>	80	
		Channel Name: EJWT						
Additional Notes								
The Panel Light and Horn will be OFF. Permissives: Comp Oil Temp and Engine Jacket Water Temp								

State 8

Label	Running								
DS Outputs									
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	☐	DS4	
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	☐	DS4	
Additional Notes									
The Panel Light will be set to solid green and the Horn will be OFF .									

State 9

Label	CoolDown	Timer:			min	30	sec	
		Timeout:			min		sec	
DS Outputs								
Output On (Beginning of State)	☐	DS1	☐	DS2	☐	DS3	☐	DS4
Output Off (End of State)	☐	DS1	☐	DS2	☐	DS3	☐	DS4
Permissive A								
☐	Enable	Channel				REL	Setpoint	
		Channel Name:						
Permissive B								
☐	Enable	Channel				REL	Setpoint	
		Channel Name:						
Post Lube								
☐	Enable	Output				Time(secs)		
Additional Notes								
The Panel Light will be set to solid red and the Horn will be OFF.								

Control Outputs (DSs)

The Control **Outputs (DSs)** are fully configured in each state via On/Off check boxes. These selections (denoted by a check mark) will either turn an output On at the beginning of the state, or Off at the end of the state. Once On is selected for a particular DS, it will remain "On" until it is turned Off at the end of a subsequent State.

Each of these discrete outputs can be configured to be Energized when Running or Energized when Stopped. Note that DS1 is fixed at Energized when Running.

An Off Delay is used to delay an output switch from engaging upon a fault condition or for a manual stop. The delay time is entered in seconds. An off delay time cannot be applied to DS1. DS1 is typically the fuel output.

OUTPUT #	OUTPUT NAME / FUNCTION	STATE	Off Delay (s)
DS1	DER to ADEM	Energized when Running	N/A
DS2	START to ADEM	Energized when Running	0
DS3	SAFETY START GAS	Energized when Running	0
DS4	COMP PRELUBE	Energized when Running	0

Digital Outputs (DOs)

The **Digital Outputs (DOs)** control various system components based on the active system state. These outputs engage and disengage control elements such as valves, alarms, and other equipment necessary for proper system operation. Each output is assigned a specific function and is activated or deactivated according to the configured system states.

Each digital output can be set to different states depending on the system sequence:

- **O (Off):** The output is deactivated.
- **X (On):** The output is activated.
- **S (Script):** A script controls the output's behavior dynamically. See **Scripted Output Functions** for additional details.
- **N/A:** This output is configurable thru the Global System screen and is not Stage related.

The digital outputs operate in coordination with the **Start-Up Sequence**, ensuring correct system operation across the ten operational states (State 0 to State 9) as well as fault and emergency shutdown (ESD) conditions.

For additional details regarding the configuration of digital outputs, refer to the **DE-4000 Operation and Configuration Manual**.

Outputs			Stage										Fault	ESD
			0	1	2	3	4	5	6	7	8	9		
Terminal Board 1	DO1	FUEL GAS VALVE	X					O				X		
	DO2	RATED RPM PERMISSIVE												
	DO3	ASC RESET												
	DO4	PROPANE COLD START												
	DO5	BATTERY SAVER												
	DO6	BATTERY SAVER (FUTURE)												
	DO7	BATTERY SAVER (FUTURE)												
	DO8	STATUS												
Terminal Board 2	DO1	BLOWDOWN VALVE												
	DO2	BYPASS VALVE												
	DO3	Not Used												
	DO4	Not Used												
	DO5	Not Used												
	DO6	Not Used												
	DO7	Not Used												
	DO8	Not Used												

Scripted Output Functions

Some **Digital Outputs (DOs)** require scripting to achieve complex or conditional functionality beyond simple on/off control. These outputs are not directly toggled by the Start-Up Sequence but instead depend on specific system conditions, logic operations, or external inputs.

Scripting allows for advanced control, such as sequencing multiple outputs based on sensor feedback, implementing interlocks, or dynamically adjusting output states based on operational parameters. Outputs configured with scripting ensure that the system responds intelligently to varying conditions, improving reliability and automation.

Blowdown & Bypass Valves

Programming based on State 2 Additional Notes

Light/Horn 1, 2, 3, A

These 4 outputs work in conjunction to produce various color lights and states, as well as an audible horn as defined in each State.

Auto-Restart / Auto-Stop — Sequence of Operations

Overview

The Auto-Restart system automatically stops and restarts a gas compressor based on suction/discharge pressure conditions, with configurable retry logic and an Auto/Local mode system for personnel safety lockout.

Configuration Parameters (Global Settings Page 11)

Parameter	Key	Description	Default
Enable Auto Restart	AutoRestartEnabled	Master enable for entire AR system	0 (off)
Restart on Any Fault	ARRestartAnyFault	Auto-restart after any fault, not just pressure	0 (off)
Max Retries	AutoRestartMaxRetries	Max restart attempts before permanent fault	3
Retry Delay (mins)	AutoRestartRetryDelay	Standby countdown before each restart attempt	60 mins
Reset Time (mins)	AutoRestartResetTime	Run time required to reset retry counter to 0	20 mins
Low Stop Input	ARSuctTerm / ARSuctChan	Suction pressure input channel	None
Stop/Standby Below	ARSuctLoStandby	Suction pressure below this → stop & standby	—
Restart Above	ARSuctHiStandby	Suction must rise above this to restart	—
High Stop Input	ARDischTerm / ARDischChan	Discharge pressure input channel	None
Stop/Standby Above	ARDischHiStandby	Discharge pressure above this → stop & standby	—
Restart Below	ARDischLoStandby	Discharge must drop below this to restart	—
Switch Input	ARAutoLocalTerm / ARAutoLocalChan	Physical auto/local selector switch	None
Default System Mode	ARPowerUpMode	Mode on power up: "auto" or "local"	"local"

Auto/Local Mode System

Purpose

Provides a personnel safety lockout mechanism. When in **Local** mode, the unit will not auto-start. Technicians set the unit to Local before working on it.

Mode Determination

The `AutoLocal` virtual variable holds the current mode (`"Auto"` or `"Local"`). It is determined by:

1. Power-Up Default

On first boot after power cycle, `AutoLocal` is set based on `ARPowerUpMode` :

- `"auto"` → unit starts in Auto mode
- `"local"` (default) → unit starts in Local mode

2. Physical Switch Input (if configured)

- **Closed (value > 0.5)** → Auto
- **Open (value ≤ 0.5)** → Local (fail-safe: broken wire = Local)
- Switch state change overrides current mode
- Switch in Local position **always forces Local**

3. Manual Stop Override

- If a **Manual Stop** (fault type 4) or **Remote Stop** (fault type 14) occurs while in Auto mode, the system automatically reverts to **Local** mode
- This prevents the unit from auto-restarting after an intentional stop
- The operator must explicitly switch back to Auto to re-enable auto-restart

Mode Evaluation Order (each tick)

1. If physical switch configured:
 - a. If switch state changed → override `AutoLocal` to match
 - b. If switch is open (Local) → force `AutoLocal` = `"Local"`
2. If faulted AND in Auto AND fault type is 4 or 14 → force Local

Standby Entry (Auto-Stop)

Trigger Conditions

The system checks pressure values every tick (via `checkAutoRestart()`). Standby is triggered when **all** of the following are true:

1. `AutoRestartEnabled == 1`
2. `AutoLocal == "Auto"`
3. Unit is running (state 1–8) or ready (state 0, not faulted)
4. Retry count < Max Retries

Low Pressure Stop (Suction)

- Suction channel configured (`ARSuctTerm` / `ARSuctChan` > 0)
- `ARSuctLoStandby` is set
- Current suction value **drops below** `ARSuctLoStandby`
- Cause: `"lowSuction"`

High Pressure Stop (Discharge)

- Discharge channel configured (`ARDischTerm` / `ARDischChan` > 0)

- ARDischHiStandby is set
- Current discharge value **rises above** ARDischHiStandby
- Cause: "highDischarge"

Auto Mode Entry (State 0)

- Unit is at state 0 (ready), not faulted, no start pending
- Both Auto mode conditions met
- This handles the case where both are in Auto and the unit is simply stopped
- Cause: "autoMode"

Standby Entry Sequence

1. Stop the unit (call stop())
2. Set state to 10 (Standby)
3. Record standby metadata:
 - standbyType: "Low" or "High" or "Auto"
 - standbyLabel: channel label + terminal:channel
 - standbyTerm / standbyCh: source channel
 - arStandbyCause: "lowSuction", "highDischarge", or "autoMode"
4. Set arFaultRetryPending = true
5. Reset debounce state
6. Log standby event (event type 8, pause-circle amber icon)
 - Includes channel value and setpoint
7. Start standby countdown timer (AutoRestartRetryDelay minutes)

Max Retries Exceeded

If retry count $\geq$ Max Retries when entering standby:

1. Set permanent fault (type 10)
 - Low: "Low <label> ARS fail SP:<setpoint>"
 - High: "High <label> ARS fail SP:<setpoint>"
 - Auto: "<type> <label> ARS fail"
2. Stop unit, set state to 0
3. Unit stays faulted until manual reset
4. Retry counter remains at max (only resets after running for ResetTime minutes)

Standby Countdown (State 10)

During Countdown

- Status string shows: "Standby <type> <label> <Xm Xs> [Auto]" or "[Local]"
- Timer counts down from AutoRestartRetryDelay minutes

Mode Change During Countdown

- If mode switches to **Local** during countdown:
 - Status shows "Standby – Local mode"
 - Countdown continues but unit will **not** restart
 - Cancels auto-restart with custom fault "– Auto-Restart Cancelled"

Timer Expiry — Pre-Start Validation

When the standby timer reaches 0:

1. Check if still in Auto mode
 - If Local → cancel, return
2. Validate restart conditions based on cause:
 - a. lowSuction:
 - Read current suction value
 - Must be ABOVE ARSuctHiStandby (Restart Above threshold)
 - If not met → show "Waiting: pressure X.X <= Y", keep waiting
 - b. highDischarge:
 - Read current discharge value
 - Must be BELOW ARDischLoStandby (Restart Below threshold)
 - If not met → show "Waiting: pressure X.X >= Y", keep waiting
 - c. autoMode:
 - Always passes (no pressure check needed during first start)
3. If conditions NOT met:
 - Reset debounce, keep waiting
 - Does NOT count as a retry attempt
 - Keeps checking each tick until conditions are met
4. If conditions ARE met:
 - Start 5-second debounce timer
 - Status: "Conditions met, restarting in Xs"
 - After 5 seconds of sustained conditions:
 - Proceed to restart

Restart Attempt

When conditions are met and debounce completes:

1. Set state to 0 (Ready)
2. Set autoReset = true, resetFlag = true (clears any fault)
3. Set startPressed = true (triggers start sequence)
4. Log auto-restart attempt (event type 9, play-circle cyan icon)
 - "Auto-Restart attempt N"
5. Increment retry counter (asRetries + 1)
6. Unit proceeds through normal start sequence (states 1 → 2 → ... → 8)

Fault During Restart Attempt

If the unit faults during the start sequence (e.g., pressure drops again mid-crank):

With "Restart on Any Fault" ENABLED (default)

1. Unit faults → goes to state 0 with fault flag
2. checkAutoRestart() sees: state 0, faulted, Auto mode, ARRestartAnyFault=1
3. Check fault type:
 - Fault type 4 (Manual Stop) or 14 (Remote Stop) → SKIP (intentional)

- Any other fault type → proceed
4. Check retry count < Max Retries
 - If yes: clear fault (resetFlag + autoReset), log event, re-enter standby cycle
 - If no: stop retrying, unit stays faulted

With "Restart on Any Fault" DISABLED

1. Unit faults → goes to state 0 with fault flag
2. checkAutoRestart() sees the fault but ARRestartAnyFault=0
3. Sets retry count to max (effectively stops retrying)
4. Unit stays faulted until manual reset

Retry Counter Reset

If the unit successfully starts and runs for `AutoRestartResetTime` minutes:

1. checkAutoRestart() checks: asRetries > 0 AND elapsedTime > ResetTime AND state > 0
2. Resets asRetries to 0
3. Full retry budget is restored for next standby event

Event Log Entries

Event Type	Icon	Color	Description
8 (Standby)	pause-circle	Amber (#e8b339)	Entering standby state, fault retry during AR cycle
9 (Auto Restart)	play-circle	Cyan (#13c2c2)	Restart attempt initiated

Dashboard Display

Status Strings (State 10)

Scenario	Status String
Countdown active, Auto mode	"Standby Low suction 4m32s [Auto]"
Timer expired, conditions not met	"Waiting: pressure 25.0 <= 30"
Timer expired, conditions met	"Conditions met, restarting in 4s"

Virtual Variables

Variable	Purpose
AutoLocal	Current mode: "Auto" or "Local"
asr / _asRetries	Current retry count (for dashboard display)

<code>_standbyLoHi</code>	0 = low, 1 = high (standby type indicator)
<code>_standbyTerm / _standbyCh</code>	Standby source channel

Complete Lifecycle Example

Scenario: Low suction pressure causes standby, then recovers

```

Time 00:00 – Unit running (state 8), Auto mode, suction = 50 PSI
              Stop Below = 40, Restart Above = 45, Retry Delay = 5 min

Time 01:00 – Suction drops to 38 PSI (below 40)
              → checkAutoRestart() detects: val(38) < ARSuctLoStandby(40)
              → enterAutoRestartStandby("lowSuction")
              → Unit stops, enters state 10
              → Log: "Standby: Low suction T1:IN5 value: 38" (type 8)
              → Countdown starts: 5 minutes

Time 01:00–06:00 – Standby countdown
                  → Status: "Standby Low suction T1:IN5 4m30s [Auto]"
                  → Suction still at 38 PSI

Time 06:00 – Timer expires, check restart conditions
              → Suction = 38 PSI, threshold = 45
              → 38 < 45 → conditions NOT met
              → Status: "Waiting: pressure 38.0 <= 45"
              → No retry consumed, keeps checking

Time 08:00 – Suction recovers to 46 PSI
              → 46 > 45 → conditions met!
              → Start 5-second debounce
              → Status: "Conditions met, restarting in 5s"

Time 08:05 – Debounce complete, suction still at 46
              → Set state 0, clear fault, press start
              → Log: "Auto-Restart attempt 1" (type 9)
              → asRetries = 1
              → Unit begins start sequence (states 1 → 2 → ... → 8)

Time 08:30 – Unit running at state 8
              → If runs for 20 min (Reset Time) → asRetries resets to 0

```

Scenario: Fault during restart, retry through fault

```

Time 00:00 – Unit in standby (state 10), countdown expired, conditions met

Time 00:05 – Debounce complete, restart attempt 1 initiated
              → Unit starts, enters state 3 (cranking)

Time 00:15 – Fault occurs during cranking (e.g., overcrank timeout, type 5)

```

- Unit faults, goes to state 0
- ARRestartAnyFault = 1 (enabled)
- Fault type 5 ≠ 4 and ≠ 14 → eligible for retry
- asRetries(1) < maxRetries(3) → clear fault
- Log: "Auto-Restart: restarting after fault (type 5)" (type 8)
- Unit returns to ready state, re-enters standby on next tick

Time 00:16 – checkAutoRestart() sees state 0, Auto, not faulted, retries < max
→ enterAutoRestartStandby("autoMode")
→ New standby countdown begins

Time 05:16 – Timer expires, conditions met (autoMode = always passes)
→ Restart attempt 2 initiated

... (repeats until success or max retries)

Time XX:XX – asRetries reaches 3 (max)
→ enterAutoRestartStandby sees retries >= max
→ Permanent fault: "Auto ARS fail"
→ Unit stays faulted, requires manual reset

Scenario: Manual stop during auto-restart cycle

Time 00:00 – Unit in standby (state 10), countdown active, Auto mode

Time 02:00 – Operator presses STOP button
→ sendStop() fires (state 10 included in stop conditions)
→ buttons() processes stop: fault type 4 (Manual Stop)
→ checkAutoLocal() sees fault type 4 while in Auto
→ Forces AutoLocal = "Local"
→ Unit will NOT auto-restart (Local mode)
→ Operator must switch back to Auto to re-enable